

Also,

$$\Pr(A_2|A_1 \cap B) = \frac{\Pr(A_1 \cap A_2 \cap B)}{\Pr(A_1 \cap B)} = \frac{\Pr(A_1 \cap A_2 \cap B)/\Pr(B)}{\Pr(A_1 \cap B)/\Pr(B)} = \frac{\Pr(A_1 \cap A_2|B)}{\Pr(A_1|B)}.$$

Hence, $\Pr(A_2|A_1 \cap B) = \Pr(A_2|B)$.

For the “if” direction, we assume that $\Pr(A_2|A_1 \cap B) = \Pr(A_2|B)$, and we must prove that A_1 and A_2 are conditionally independent given B . That is, we must prove that $\Pr(A_1 \cap A_2|B) = \Pr(A_1|B) \Pr(A_2|B)$. We know that

$$\Pr(A_1 \cap A_2|B) \Pr(B) = \Pr(A_2|A_1 \cap B) \Pr(A_1 \cap B),$$

since both sides are equal to $\Pr(A_1 \cap A_2 \cap B)$. Divide both sides of this equation by $\Pr(B)$ and use the assumption $\Pr(A_2|A_1 \cap B) = \Pr(A_2|B)$ together with $\Pr(A_1 \cap B)/\Pr(B) = \Pr(A_1|B)$ to obtain

$$\Pr(A_1 \cap A_2|B) = \Pr(A_2|B) \Pr(A_1|B).$$

23. (a) Conditional on B the events $A_1, \dots, A_{11}$ are independent with probability 0.8 each. The conditional probability that a particular collection of eight programs out of the 11 will compile is $0.8^8 0.2^3 = 0.001342$. There are $\binom{11}{8} = 165$ different such collections of eight programs out of the 11, so the probability of exactly eight programs will compile is $165 \times 0.001342 = 0.2215$.
- (b) Conditional on B^c the events $A_1, \dots, A_{11}$ are independent with probability 0.4 each. The conditional probability that a particular collection of eight programs out of the 11 will compile is $0.4^8 0.6^3 = 0.0001416$. There are $\binom{11}{8} = 165$ different such collections of eight programs out of the 11, so the probability of exactly eight programs will compile is $165 \times 0.0001416 = 0.02335$.
24. Let $n > 1$, and assume that $A_1, \dots, A_n$ are mutually exclusive. For the “if” direction, assume that at most one of the events has strictly positive probability. Then, the intersection of every collection of size 2 or more has probability 0. Also, the product of every collection of 2 or more probabilities is 0, so the events satisfy Definition 2.2.2 and are mutually independent. For the “only if” direction, assume that the events are mutually independent. The intersection of every collection of size 2 or more is empty and must have probability 0. Hence the product of the probabilities of every collection of size 2 or more must be 0. This means that at least one factor from every product of at least 2 probabilities must itself be 0. Hence there can be no more than one of the probabilities greater than 0, otherwise the product of the two nonzero probabilities would be nonzero.

2.3 Bayes’ Theorem

Commentary

This section ends with two extended discussions on how Bayes’ theorem is applied. The first involves a sequence of simple updates to the probability of a specific event. It illustrates how conditional independence allows one to use posterior probabilities after observing some events as prior probabilities before observing later events. This idea is subtle, but very useful in Bayesian inference. The second discussion builds upon this idea and illustrates the type of reasoning that can be used in real inference problems. Examples 2.3.7 and 2.3.8 are particularly useful in this regard. They show how data can bring very different prior probabilities into

closer posterior agreement. Exercise 12 illustrates the effect of the size of a sample on the degree to which the data can reduce differences in subjective probabilities.

Statistical software like *R* can be used to facilitate calculations like those that occur in the above-mentioned examples. For example, suppose that the 11 prior probabilities are assigned to the vector `prior` and that the data consist of `s` successes and `f` failures. Then the posterior probabilities can be computed by

```
ints=1:11
post=prior*((ints-1)/10)^s*(1-(ints-1)/10)^f
post=post/sum(post)
```

Solutions to Exercises

1. It must be true that $\sum_{i=1}^k \Pr(B_i) = 1$ and $\sum_{i=1}^k \Pr(B_i | A) = 1$. However, if $\Pr(B_1 | A) < \Pr(B_1)$ and $\Pr(B_i | A) \leq \Pr(B_i)$ for $i = 2, \dots, k$, we would have $\sum_{i=1}^k \Pr(B_i | A) < \sum_{i=1}^k \Pr(B_i)$, a contradiction. Therefore, it must be true that $\Pr(B_i | A) > \Pr(B_i)$ for at least one value of i ($i = 2, \dots, k$).
2. It was shown in the text that $\Pr(A_2 | B) = 0.26 < \Pr(A_2) = 0.3$. Similarly,

$$\Pr(A_1 | B) = \frac{(0.2)(0.01)}{(0.2)(0.01) + (0.3)(0.02) + (0.5)(0.03)} = 0.09.$$

Since $\Pr(A_1) = 0.2$, we have $\Pr(A_1 | B) < \Pr(A_1)$. Furthermore,

$$\Pr(A_3 | B) = \frac{(0.5)(0.03)}{(0.2)(0.01) + (0.3)(0.02) + (0.5)(0.03)} = 0.65.$$

Since $\Pr(A_3) = 0.5$, we have $\Pr(A_3 | B) > \Pr(A_3)$.

3. Let C denote the event that the selected item is nondefective. Then

$$\Pr(A_2 | C) = \frac{(0.3)(0.98)}{(0.2)(0.99) + (0.3)(0.98) + (0.5)(0.97)} = 0.301.$$

Commentary: It should be noted that if the selected item is observed to be defective, the probability that the item was produced by machine M_2 is decreased from the prior value of 0.3 to the posterior value of 0.26. However, if the selected item is observed to be nondefective, this probability changes very little, from a prior value of 0.3 to a posterior value of 0.301. In this example, therefore, obtaining a defective is more informative than obtaining a nondefective, but it is much more probable that a nondefective will be obtained.

4. The desired probability $\Pr(\text{Cancer} | \text{Positive})$ can be calculated as follows:

$$\begin{aligned} & \frac{\Pr(\text{Cancer}) \Pr(\text{Positive} | \text{Cancer})}{\Pr(\text{Cancer}) \Pr(\text{Positive} | \text{Cancer}) + \Pr(\text{No Cancer}) \Pr(\text{Positive} | \text{No Cancer})} \\ &= \frac{(0.00001)(0.95)}{(0.00001)(0.95) + (0.99999)(0.05)} = 0.00019. \end{aligned}$$

Commentary: It should be noted that even though this test provides a correct diagnosis 95 percent of the time, the probability that a person has this type of cancer, given that he has a positive reaction to

the test, is *not* 0.95. In fact, as this exercise shows, even though a person has a positive reaction, the probability that she has this type of cancer is still only 0.00019. In other words, the probability that the person has this type of cancer is 19 times larger than it was before he took this test $\left(\frac{0.00019}{0.000001} = 19\right)$, but it is still very small because the disease is so rare in the population.

5. The desired probability $\Pr(\text{Lib.}|\text{NoVote})$ can be calculated as follows:

$$\begin{aligned} & \frac{\Pr(\text{Lib.}) \Pr(\text{NoVote}|\text{Lib.})}{\Pr(\text{Cons.}) \Pr(\text{NoVote}|\text{Cons.}) + \Pr(\text{Lib.}) \Pr(\text{NoVote}|\text{Lib.}) + \Pr(\text{Ind.}) \Pr(\text{NoVote}|\text{Ind.})} \\ &= \frac{(0.5)(0.18)}{(0.3)(0.35) + (0.5)(0.18) + (0.2)(0.50)} = \frac{18}{59}. \end{aligned}$$

6. (a) Let A_1 denote the event that the machine is adjusted properly, let A_2 denote the event that it is adjusted improperly, and let B be the event that four of the five inspected items are of high quality. Then

$$\begin{aligned} \Pr(A_1 | B) &= \frac{\Pr(A_1) \Pr(B | A_1)}{\Pr(A_1) \Pr(B | A_1) + \Pr(A_2) \Pr(B | A_2)} \\ &= \frac{(0.9) \binom{5}{4} (0.5)^5}{(0.9) \binom{5}{4} (0.5)^5 + (0.1) \binom{5}{4} (0.25)^4 (0.75)} = \frac{96}{97}. \end{aligned}$$

- (b) The prior probabilities before this additional item is observed are the values found in part (a): $\Pr(A_1) = 96/97$ and $\Pr(A_2) = 1/97$. Let C denote the event that the additional item is of medium quality. Then

$$\Pr(A_1 | C) = \frac{\frac{96}{97} \cdot \frac{1}{2}}{\frac{96}{97} \cdot \frac{1}{2} + \frac{1}{97} \cdot \frac{3}{4}} = \frac{64}{65}.$$

7. (a) Let π_i denote the posterior probability that coin i was selected. The prior probability of each coin is $1/5$. Therefore

$$\pi_i = \frac{\frac{1}{5} p_i}{\sum_{j=1}^5 \frac{1}{5} p_j} \quad \text{for } i = 1, \dots, 5.$$

The five values are $\pi_1 = 0$, $\pi_2 = 0.1$, $\pi_3 = 0.2$, $\pi_4 = 0.3$, and $\pi_5 = 0.4$.

- (b) The probability of obtaining another head is equal to

$$\sum_{i=1}^5 \Pr(\text{Coin } i) \Pr(\text{Head} | \text{Coin } i) = \sum_{i=1}^5 \pi_i p_i = \frac{3}{4}.$$

- (c) The posterior probability π_i of coin i would now be

$$\pi_i = \frac{\frac{1}{5} (1 - p_i)}{\sum_{j=1}^5 \frac{1}{5} (1 - p_j)} \quad \text{for } i = 1, \dots, 5.$$

Thus, $\pi_1 = 0.4, \pi_2 = 0.3, \pi_3 = 0.2, \pi_4 = 0.1$, and $\pi_5 = 0$. The probability of obtaining a head on the next toss is therefore $\sum_{i=1}^5 \pi_i p_i = \frac{1}{4}$.

8. (a) If coin i is selected, the probability that the first head will be obtained on the fourth toss is $(1 - p_i)^3 p_i$. Therefore, the posterior probability that coin i was selected is

$$\pi_i = \frac{\frac{1}{5}(1 - p_i)^3 p_i}{\sum_{j=1}^5 \frac{1}{5}(1 - p_j)^3 p_j} \quad \text{for } i = 1, \dots, 5.$$

The five values are $\pi_1 = 0, \pi_2 = 0.5870, \pi_3 = 0.3478, \pi_4 = 0.0652$, and $\pi_5 = 0$.

- (b) If coin i is used, the probability that exactly three additional tosses will be required to obtain another head is $(1 - p_i)^2 p_i$. Therefore, the desired probability is

$$\sum_{i=1}^5 \pi_i (1 - p_i)^2 p_i = 0.1291.$$

9. We shall continue to use the notation from the solution to Exercise 14 in Sec. 2.1. Let C be the event that exactly one out of seven observed parts is defective. We are asked to find $\Pr(B_j|C)$ for $j = 1, 2, 3$. We need $\Pr(C|B_j)$ for each j . Let A_i be the event that the i th part is defective. For all i , $\Pr(A_i|B_1) = 0.02$, $\Pr(A_i|B_2) = 0.1$, and $\Pr(A_i|B_3) = 0.3$. Since the seven parts are conditionally independent given each state of the machine, the probability of each possible sequence of seven parts with one defective is $\Pr(A_i|B_j)[1 - \Pr(A_i|B_j)]^6$. There are seven distinct such sequences, so

$$\Pr(C|B_1) = 7 \times 0.02 \times 0.98^6 = 0.1240,$$

$$\Pr(C|B_2) = 7 \times 0.1 \times 0.9^6 = 0.3720,$$

$$\Pr(C|B_3) = 7 \times 0.3 \times 0.7^6 = 0.2471.$$

The expression in the denominator of Bayes' theorem is

$$\Pr(C) = 0.8 \times 0.1240 + 0.1 \times 0.3720 + 0.1 \times 0.2471 = 0.1611.$$

Bayes' theorem now says

$$\Pr(B_1|C) = \frac{0.8 \times 0.1240}{0.1611} = 0.6157,$$

$$\Pr(B_2|C) = \frac{0.1 \times 0.3720}{0.1611} = 0.2309,$$

$$\Pr(B_3|C) = \frac{0.1 \times 0.2471}{0.1611} = 0.1534.$$

10. Bayes' theorem says that the posterior probability of each B_i is $\Pr(B_i|E) = \Pr(B_i) \Pr(E|B_i) / \Pr(E)$. So $\Pr(B_i|E) < \Pr(B_i)$ if and only if $\Pr(E|B_i) < \Pr(E)$. Since $\Pr(E) = 3/4$, we need to find those i for which $\Pr(E|B_i) < 3/4$. These are $i = 5, 6$.
11. This time, we want $\Pr(B_4|E^c)$. We know that $\Pr(E^c) = 1 - \Pr(E) = 1/4$ and $\Pr(E^c|B_4) = 1 - \Pr(E|B_4) = 1/4$. This means that E^c and B_4 are independent so that $\Pr(B_4|E^c) = \Pr(B_4) = 1/4$.

12. We are doing the same calculations as in Examples 2.3.7 and 2.3.8 except that we only have five patients and three successes. So, in particular

$$\Pr(B_j | A) = \frac{\Pr(B_j) \binom{5}{3} ([j-1]/10)^3 (1 - [j-1]/10)^2}{\sum_{i=1}^{11} \Pr(B_i) \binom{5}{3} ([i-1]/10)^3 (1 - [i-1]/10)^2}. \quad (\text{S.2.5})$$

In one case, $\Pr(B_i) = 1/11$ for all i , and in the other case, the prior probabilities are given in the table in Example 2.3.8 of the text. The numbers that show up in both calculations are

i	$\left(\frac{i-1}{10}\right)^3 \left(1 - \frac{i-1}{10}\right)^2$	i	$\left(\frac{i-1}{10}\right)^3 \left(1 - \frac{i-1}{10}\right)^2$
1	0	7	0.0346
2	0.0008	8	0.0309
3	0.0051	9	0.0205
4	0.0132	10	0.0073
5	0.0230	11	0
6	0.0313		

We can use these with the two sets of prior probabilities to compute the posterior probabilities according to Eq. (S.2.5).

i	Example 2.3.7	Example 2.3.8	i	Example 2.3.7	Example 2.3.8
1	0	0	7	0.2074	0.1641
2	0.0049	0.0300	8	0.1852	0.0879
3	0.0307	0.0972	9	0.1229	0.0389
4	0.07939	0.1633	10	0.0437	0.0138
5	0.1383	0.1969	11	0	0
6	0.1875	0.2077			

These numbers are not nearly so close as those in the examples in the text because we do not have as much information in the small sample of five patients.

13. (a) Let B_1 be the event that the coin is fair, and let B_2 be the event that the coin has two heads. Let H_i be the event that we obtain a head on the i th toss for $i = 1, 2, 3, 4$. We shall apply Bayes' theorem conditional on $H_1 H_2$.

$$\begin{aligned} & \Pr(B_1 | H_1 \cap H_2 \cap H_3) \\ &= \frac{\Pr(B_1 | H_1 \cap H_2) \Pr(H_3 | B_1 \cap H_1 \cap H_2)}{\Pr(B_1 | H_1 \cap H_2) \Pr(H_3 | B_1 \cap H_1 \cap H_2) + \Pr(B_2 | H_1 \cap H_2) \Pr(H_3 | B_2 \cap H_1 \cap H_2)} \\ &= \frac{(1/5) \times (1/2)}{(1/5) \times (1/2) + (4/5) \times 1} = \frac{1}{9}. \end{aligned}$$

- (b) If the coin ever shows a tail, it can't have two heads. Hence the posterior probability of B_1 becomes 1 after we observe a tail.

14. In Exercise 23 of Sec. 2.2, B is the event that the programming task was easy. In that exercise, we computed $\Pr(A|B) = 0.2215$ and $\Pr(A|B^c) = 0.02335$. We are also told that $\Pr(B) = 0.4$. Bayes'

theorem tells us that

$$\begin{aligned}\Pr(B|A) &= \frac{\Pr(B)\Pr(A|B)}{\Pr(B)\Pr(A|B) + \Pr(B^c)\Pr(A|B^c)} = \frac{0.4 \times 0.2215}{0.4 \times 0.2215 + (1 - 0.4)0.02335} \\ &= 0.8635.\end{aligned}$$

15. The law of total probability tells us how to compute $\Pr(E_1)$.

$$\Pr(E_1) = \sum_{i=1}^{11} \Pr(B_i) \frac{i-1}{10}.$$

Using the numbers in Example 2.3.8 for $\Pr(B_i)$ we obtain 0.274. This is smaller than the value 0.5 computed in Example 2.3.7 because the prior probabilities in Example 2.3.8 are much higher for the B_i with low values of i , particularly $i = 2, 3, 4$, and they are much smaller for those B_i with large values of i . Since $\Pr(E_1)$ is a weighted average of the values $(i-1)/10$ with the weights being $\Pr(B_i)$ for $i = 1, \dots, 11$, the more weight we give to small values of $(i-1)/10$, the smaller the weighted average will be.

16. (a) From the description of the problem $\Pr(D_i|B) = 0.01$ for all i . If we can show that $\Pr(D_i|B^c) = 0.01$ for all i , then $\Pr(D_i) = 0.01$ for all i . We will prove this by induction. We have assumed that D_1 is independent of B and hence it is independent of B^c . This makes $\Pr(D_1|B^c) = 0.01$. Now, assume that $\Pr(D_i|B^c) = 0.01$ for all $i \leq j$. Write

$$\Pr(D_{j+1}|B^c) = \Pr(D_{j+1}|D_j \cap B^c) \Pr(D_j|B^c) + \Pr(D_{j+1}|D_j^c \cap B^c) \Pr(D_j^c|B^c).$$

The induction hypothesis says that $\Pr(D_j|B^c) = 0.01$ and $\Pr(D_j^c|B^c) = 0.99$. In the problem description, we have $\Pr(D_{j+1}|D_j \cap B^c) = 2/5$ and $\Pr(D_{j+1}|D_j^c \cap B^c) = 1/165$. Plugging these into (16a) gives

$$\Pr(D_{j+1}|B^c) = \frac{2}{5} \times 0.01 + \frac{1}{165} \times 0.99 = 0.01.$$

This completes the proof.

- (b) It is straightforward to compute

$$\Pr(E|B) = 0.99 \times 0.99 \times 0.01 \times 0.01 \times 0.99 \times 0.99 = 0.00009606.$$

By the conditional independence assumption stated in the problem description, we have

$$\Pr(E|B^c) = \Pr(D_1^c|B^c) \Pr(D_2^c|D_1^c \cap B^c) \Pr(D_3|D_2^c \cap B^c) \Pr(D_4|D_3 \cap B^c) \Pr(D_5^c|D_4 \cap B^c) \Pr(D_6^c|D_5^c \cap B^c).$$

The six factors on the right side of this equation are respectively 0.99, $164/165$, $1/165$, $2/5$, $3/5$, and $164/165$. The product is 0.001423. It follows that

$$\begin{aligned}\Pr(B|E) &= \frac{\Pr(E|B)\Pr(B)}{\Pr(E|B)\Pr(B) + \Pr(E|B^c)\Pr(B^c)} \\ &= \frac{0.00009606 \times (2/3)}{0.00009606 \times (2/3) + 0.001423 \times (1/3)} = 0.1190.\end{aligned}$$